

WILL SUAN

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EDUCATION

The University of Texas at Austin

B.S. Mathematics, Certificate in Scientific Computation & Data Sciences

May 2026

GPA: 3.82

Research Interests: neural population dynamics, representation geometry, latent-state modeling, structure-preserving numerical methods

EXPERIENCE

Computational Visualization Center

Research Assistant

Dec 2025 – May 2026

Austin, TX

- Built controlled Hamiltonian optimization modules for GRL-SNAM, a geometric RL project on navigation in unknown environments.
- Benchmarked 2D navigation policies against PF, CBF, A*, and deep RL, measuring SPL, clearance, and noise robustness.
- Built and tested PyTorch modules for PHAST, a port-Hamiltonian architecture for forecasting dissipative dynamical systems.
- Investigated parameter identifiability — when the model forecasts well vs. when it recovers true damping and mass fields.

Saronic Technologies

Machine Learning Intern

May 2025 – Aug 2025

Austin, TX

- Profiled class balance and annotation quality across 50,000+ maritime images, flagging gaps for targeted data collection.
- Built a maritime detection and tracking prototype for Saronic's autonomous surface vessels, profiling inference on edge hardware.
- Wrote evaluation scripts computing precision, recall, F1, and identity-switch rate across vessel classes, sea states, and lighting.
- Ran hyperparameter sweeps that achieved a 30% latency reduction at fixed accuracy, fitting the onboard compute budget.

Center for Theoretical and Computational Neuroscience

Research Assistant

Jan 2024 – Mar 2024

Austin, TX

- Simulated finite-size spiking networks to test replica-mean-field (RMF) predictions for pair correlations and firing rates.
- Implemented point-process intensity models in Python, tracking transition rates between metastable states across coupling regimes.
- Fit posterior distributions over network parameters using MCMC, validating recovery against ground-truth synthetic data.

Lockheed Martin

Research Intern

Jun 2023 – Dec 2023

Fort Worth, TX / Remote

- Wrote finite-difference time-domain simulations in C++ modeling magnetic-field effects on hypersonic plasma-sheath density.
- Quantified RF attenuation across 12 magnet geometries at Mach 5–8, identifying 3 that opened a communication window.
- Validated simulations against analytical plasma models and published experimental data, documenting convergence for team reference.
- Ran MPI-parallelized sweeps over plasma density and magnet strength, mapping the sensitivity of the communication-window envelope.
- Co-authored an internal technical report on magnet-geometry trade-offs, used by the aerodynamics team for design reviews.

Optimal

Data Science Intern

Jun 2022 – Aug 2022

Irvine, CA

- Built ETL pipelines integrating Microsoft, Mazda, and Kroger campaign data into the agency's shared customer data warehouse.
- Applied k-means clustering and regression to segment audience profiles, presenting recommendations to the campaign strategy team.
- Automated weekly reporting dashboards tracking campaign KPIs, A/B test significance, and ROI across 3 concurrent clients.

LEADERSHIP

Austin Bat Cave — Volunteer

Sep 2024 – May 2026

- Volunteered weekly as a one-on-one writing tutor for Title 1 students ages 6–18, supporting drafting from prompt to finished piece.
- Mentored students through college essays and submissions for Austin Bat Cave's annual anthology across four semesters.
- Trained incoming volunteer tutors in Bat Cave's one-on-one approach, pairing them with returning students during their first semesters.

Society of Industrial and Applied Mathematics — Chapter Officer

Sep 2024 – May 2026

- Organized guest speakers from academia and national labs; secured funding for 24 students to compete in COMAP MCM/ICM.
- Led weekly reading group on scientific machine learning, covering data assimilation, physics-informed networks, and surrogate modeling.

Texas Rocket Engineering Lab — Avionics Lead

Jan 2023 – Feb 2024

- Led 15 students across wiring, power, RF, and ground-antenna systems, coordinating HIL testing with the propulsion and GNC teams.
- Designed tracking-antenna pointing and DSP pipelines for telemetry recovery during a 35 km launch vehicle's flight envelope.

TECHNICAL SKILLS

Languages: Python, R, Julia, C/C++ (MPI, OpenMP), CUDA, SQL

Methods: numerical linear algebra, convex optimization, MCMC, point processes, reinforcement learning, finite-difference methods

Graduate Coursework: Numerical Analysis, Complex Analysis, Tensor Methods, Physical Simulation for Computer Graphics, Computational Physics, Multidimensional Data Analysis, Applied Data Analysis & ML, Quantum Algorithms for Scientific Computing